

Doppler Weather Radar (DWR) Operations and Maintenance

Unit 1: Principles of Radar Meteorology & the IMD DWR Network

Trainer notes — for internal training use on the Capacity Connect platform.

Learning Objectives

- Explain the basic principle of how weather radar detects precipitation
- Describe the layout and purpose of IMD's DWR network
- Understand the Doppler effect's role in measuring radial velocity

How Weather Radar Works

A weather radar transmits pulses of microwave energy and listens for the fraction reflected back by precipitation particles. The strength of the returned signal (reflectivity) relates to the size and concentration of the particles, while the time delay of the return indicates the target's range from the radar.

The Doppler Effect

A Doppler Weather Radar goes further, measuring the shift in frequency between the transmitted and received pulse caused by the motion of precipitation particles toward or away from the radar - the same Doppler effect that changes the pitch of a passing siren. This shift is converted into radial velocity, showing whether air (carrying precipitation) is moving toward or away from the radar, which is critical for detecting rotation within severe storms.

IMD's DWR Network

IMD operates a network of S-band and C-band Doppler radars strategically located along the coasts (for cyclone monitoring) and across the interior (for general severe weather and monsoon monitoring), each covering a radius of roughly 250-500 km depending on band and configuration. Overlapping coverage between adjacent radars provides redundancy and improves tracking of systems as they move between coverage areas.

Summary

Every radar product used later in this course - reflectivity, velocity, and derived nowcasting guidance - traces back to these two basic measurements: signal strength and frequency shift.